

CSCE 632 ACCESSIBLE COMPUTING · MODULE 02 · LECTURE NOTES

The Range

Permanent, temporary, situational — and the same requirement in all three.

MODULE OUTCOMES

Explain the permanent-temporary-situational framing.

State what is known about prevalence, and its limits.

Describe the assistive technology landscape.

Explain why the population is larger than it appears.

Use the framing to argue for a requirement.

1 Three durations

The constraint	Permanent	Temporary	Situational
Use of one hand only	Amputation, hemiplegia	A broken arm, a recent injury	Holding a child, a bag, a handrail
Cannot see the screen	Blindness	Dilated pupils after an eye examination	Bright sunlight, a cracked display, driving
Cannot hear the audio	Deafness	An ear infection	A loud bar, a quiet library, no headphones
Cannot speak	Non-speaking, laryngectomy	Laryngitis	A quiet office, a meeting, a baby asleep
Reduced attention or memory	ADHD, some cognitive conditions	Concussion, severe sleep deprivation, grief	Exhausted, distracted, doing three things

The design requirement is identical across each row — which is the whole point of the framing, and is why it persuades people the compliance argument does not: it connects the requirement to something the listener has experienced. **This table is the module's single most useful artefact**, and it is worth being able to produce a row of it for any constraint you are asked to justify.

WHICH MEANS THE SITUATIONAL COLUMN IS THE PERSUASIVE ONE AND THE PERMANENT COLUMN IS THE OBLIGATORY ONE

The situational cases make the requirement feel relevant — everybody has tried to use a phone one-handed in bright sunlight, and that is a real design argument rather than a rhetorical trick, because the interface genuinely does fail in both cases for the same reason.

And the permanent cases are why the requirement is obligatory — a situational constraint is an inconvenience and a permanent one is exclusion, which are not the same thing and should not be presented as though they were.

So use the situational framing to get agreement and the permanent one to set the standard — because the situational case tolerates a workaround ('wait until you are indoors') and the permanent one does not, and a requirement argued only from the situational case will be met only to the situational standard.

Which is Module 01 §2's limit restated: the broad benefit is real and is not the justification, and conflating them leads directly to dropping whatever lacks a mainstream beneficiary — which is most of the screen reader requirements.

2 Prevalence

- **Roughly one person in six lives with a significant disability,** by the usual international estimates — **which makes this the largest minority there is, and it is one anybody can join,** which no other minority is.
- **But the figure depends entirely on the definition — self-report, functional assessment, and administrative benefit eligibility give very different numbers, and the three are measuring different things:** identity, capability, and entitlement.
- **And it rises steeply with age, so the population is growing wherever the population is ageing — which is a demographic fact rather than a projection,** and makes this a requirement that becomes more rather than less relevant over a career.
- **Plus the undercount: people who have stopped trying to use an inaccessible system do not appear in its analytics —** which is CSCE 676 Module 08's survivorship structure operating on your own usage data.
- **So the honest claim is 'a large minority, undercounted' — and the precise number is not the argument anyway** (§4's closing callout).

AND THE ANALYTICS ARGUMENT IS WORTH REFUSING EXPLICITLY

'Our analytics show almost no screen reader users' is offered, in good faith and frequently, as evidence that the work is not needed — and **it is evidence of the opposite.**

Because the measurement is conditional on being able to use the product: the excluded are absent from the data by construction, which is **exactly CSCE 676 Module 08's survivorship structure** — you are measuring the people who got through.

And screen reader use is frequently not detectable at all — for good privacy reasons, since a detectable assistive technology is a disclosed disability — **so the absence of a signal is not even weak evidence** of absence.

So the correct response is to name the sampling problem, which is a technical objection rather than an ethical appeal — and **technical objections carry further in engineering meetings,** which is a practical fact worth using.

3 The technology

Technology	What it does	What it needs from your code
Screen reader	Speaks the interface, or sends it to a refreshable braille display.	Correct semantics: name, role, value, state (Module 08).
Screen magnifier	Enlarges a region, so only part of the screen is visible at once.	Reflow at high zoom, and no dependency between distant parts of the screen.
Switch access and scanning	One or two physical inputs, selecting sequentially from a moving highlight.	Few steps, no timeouts, and a sensible focus order (Module 05 §3).
Voice control	Operates the interface by spoken command.	Visible, speakable, matching names — the label you show must be the name it responds to.
Alternative keyboards and pointers	Different physical layouts, head pointers, eye tracking.	Full keyboard operability, and generous targets.
Captions, transcripts, and sign interpretation	Text or signed alternatives for audio content.	Accurate, synchronised text (Module 04).

Every one of these consumes your semantics rather than your pixels — a screen reader reads a tree of roles and names, voice control matches spoken words against accessible names, and switch access walks a focus order — **which is why Module 08's accessibility tree is the mechanism the whole course depends on. The semantics-not-pixels point is the bridge to Module 08**, and it is the reason a visually perfect interface can be completely inoperable.

4 Using the framing

- **Name the capability, not the condition** (Module 01 §3) — **which keeps it a design requirement rather than an accommodation request**, and those are routed through different processes with very different outcomes.
- **Give the situational case first**, because **it is the one your audience has personally experienced** and therefore the one that gets agreement.
- **Then the permanent case, as the reason the workaround is not sufficient** — which is where the standard actually comes from (§1's callout).
- **Then the test** — **which takes minutes and converts an argument into an observation**, and **is the move this whole semester is built on** (CSCE 671 Module 13 §4, CSCE 679 Module 13 §4).
- **And refuse the analytics argument by naming the sampling (§2), not by appealing to fairness. Run the test in the meeting: unplugging a mouse takes ten seconds and ends the discussion faster than any argument** — **converting the argument into an observation is the transferable skill here.**

AND THE HONEST SUMMARY

The population is large, undercounted, growing, and one anybody can join temporarily or permanently — which is a reasonable thing to say and does not depend on a precise figure, and is therefore defensible.

And the requirement does not actually depend on the count at all — an interface that excludes one user in a thousand is still defective, and the count is a prioritisation input rather than a justification.

Which is worth being clear about, because arguing from the numbers invites arguing about the numbers — and that is not the argument you want, since the numbers are genuinely uncertain (§2) and the requirement is not.

So the framing's real use is making the requirement legible to people who have not thought about it — and then the capability test settles it, which is the pattern for the next four modules: a constraint, a requirement, and a test you can run today.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Microsoft Inclusive Design (free)	§1's table in its original form — and the version worth showing to somebody who is not convinced.
The WHO World Report on Disability (free)	§2's figures , with the definitional problems stated openly — which is the part to read.
The WebAIM screen reader user surveys (free)	§3 — what people actually use, repeated over many years, and the only good public data on the question.
The W3C's How People with Disabilities Use the Web (free)	§3 with scenarios — the technologies shown in use rather than described.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Add three rows to §1's table** for constraints it does not cover.
- 2 Take one requirement** and write its situational and permanent arguments separately.
- 3 Find three published prevalence figures** and account for the difference.
- 4 Write out the survivorship argument** against the analytics claim, in four sentences.

- 5 **Install a screen reader** and complete one real task with it.
- 6 **Try your platform's voice control** on one of your own interfaces.
- 7 **Try switch or scanning access**, if your platform offers it.
- 8 **Count the steps** required for one common task in your interface.
- 9 **Check whether your visible labels match your accessible names.**
- 10 **Make the case for one requirement** in four sentences, using §4's order.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Give three rows of the durations table.
- 2 Which column persuades, which obligates, and why distinguish them?
- 3 What is the approximate prevalence, and why is the figure uncertain?
- 4 Why is the population growing?
- 5 State the undercount and its structure.
- 6 Why is the analytics argument backwards, and how do you answer it?
- 7 Name six assistive technologies and what each needs.
- 8 What do they all consume, and what does that imply?
- 9 Give the five steps for making the case.
- 10 Why should the requirement not be argued from the count?